USSS Exam — The Mythology with Examples

Read this through once. Every concept has a real example. Tuesday 2026-04-22.

The secret: this exam tests if you can FOLLOW RULES like a robot. Not reason like a human.

PART 1 — Logic-Based Reasoning (LBR)

LBR is ~30% of the exam and where most people bomb. Learn this one mental tool and you'll clear 80% of LBR questions cold.

The Core Tool: If P, Then Q

> **Every LBR question contains one or more "if-then" rules.** Your job: translate the English rule into `P → Q` and keep track of which way the arrow points.

EXAMPLE RULE

"If an agent carries a weapon, they must have passed qualification."

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P = agent carries a weapon
Q = passed qualification
Rule: P → Q
```

The arrow only goes one direction. P forces Q. But Q does NOT force P.

Move #1: MODUS PONENS (the obvious one)

> Rule: **P → Q**. Given: **P is true**. Conclusion: **Q is true**.

WORKED EXAMPLE

Rule: "If an agent carries a weapon, they passed qualification."
Given: "Agent Reyes is carrying a weapon."
Conclusion: Agent Reyes passed qualification. ✓

This one is easy. You see P, you claim Q. Done.

Move #2: MODUS TOLLENS (the contrapositive — this is THE most tested pattern)

> Rule: **P → Q**. Given: **Q is NOT true**. Conclusion: **P is NOT true**.

WORKED EXAMPLE

Rule: "If an agent carries a weapon, they passed qualification."
Given: "Agent Kim did NOT pass qualification."
Conclusion: Agent Kim is NOT carrying a weapon. ✓

Why it works: If P always forces Q, then seeing "no Q" proves "no P" — because if P were true, Q would have to be true too. It isn't, so P can't be.

This is the backbone of most "hard" LBR questions. When you see a question where a rule is given and then you're told the CONSEQUENT is false, you negate the ANTECEDENT.

TRAP #1: Affirming the Consequent (wrong direction)

✗ TRAP **Rule:** P → Q. **Given:** Q is true. **WRONG conclusion:** "Therefore P is true."

Example: "If it rains, the street is wet. The street IS wet. Therefore it rained." — WRONG. A sprinkler could have wet the street. A water main break. A spilled drink.

The arrow only points one way. Q being true does NOT mean P was the cause.

TRAP #2: Denying the Antecedent (also wrong direction)

✗ TRAP **Rule:** P → Q. **Given:** P is NOT true. **WRONG conclusion:** "Therefore Q is NOT true."

Example: "If it rains, the street is wet. It did NOT rain. Therefore the street is NOT wet." — WRONG. The street could still be wet from any other cause.

Same lesson: P not happening doesn't stop Q from happening from some other route.

The LBR Translation Table (the language they use)

The test doesn't say "If P, then Q." It uses English phrasing. You need to translate.

English phrase	Means	Direction of arrow
"If A, then B"	A → B	A forces B
"All A are B"	A → B	A forces B

"Every A is B"	A → B	A forces B
"A requires B"	A → B	A forces B
"Only B can do A"	A → B (B is necessary)	A forces B
"A only if B"	A → B	A forces B
"No A is B"	A → NOT B	A forces "not B"
"A if and only if B"	A ↔ B	Works both ways
"A unless B"	NOT B → A	"No B" forces A

"Only" Is The Word That Flips You Up

THE "ONLY" TRICK

English: "Only vehicles with a red placard may park in the VIP lot."
Common mistake: reading this as "red placard → VIP lot." WRONG.
Correct: "parked in VIP lot → has red placard." The red placard is NECESSARY.

Rule of thumb: Whatever comes AFTER "only" is the necessary condition. It goes on the RIGHT side of the arrow.

Question: "Vehicle X was parked in the VIP lot at 9:30. What can you conclude?"
Answer: Vehicle X must have a red placard. (Modus ponens: in VIP lot → has red placard. X is in VIP lot. Therefore X has placard.)

"If and only if" — Both Directions

BICONDITIONAL EXAMPLE

English: "The vault unlocks if and only if both biometric and keycard verification pass."
Translation: Unlock ↔ (Biometric AND Keycard). Both sides force each other.

Given: "The vault did NOT unlock Monday."
Conclusion: At least ONE of the two checks failed. (Not necessarily both — just at least one.)

Watch out: The trap answer says "both failed." You can't conclude that. Either failing one is enough to block unlock.

"Unless" — The Weirdest One

UNLESS EXAMPLE

English: "You can't enter the secure area unless you have a badge."
Translation: NOT badge → NOT enter. OR equivalently: Enter → has badge.

Given: "Marcus entered the secure area."
Conclusion: Marcus has a badge.

Chain Conditionals (advanced trick)

CHAIN EXAMPLE

Rule 1: "If Torres is on motorcade, then Yu is on site advance." (Torres → Yu)
Rule 2: "If Yu is on site advance, then Kim is at command post." (Yu → Kim)
Given: "Kim is NOT at command post." (NOT Kim)

Walkthrough:

```
Step 1. Chain: Torres → Yu → Kim (so Torres → Kim)
Step 2. Contrapositive of the whole chain: NOT Kim → NOT Torres
Step 3. We're told NOT Kim, so: NOT Torres. Torres is NOT on motorcade.
```

This is modus tollens across a chain. When you see a chain and the END is negated, the BEGINNING is negated too.

The Quantifier Trap: "Most" and "Some" Tell You NOTHING About Individuals

QUANTIFIER EXAMPLE

Rule: "Most field agents have completed language training."
Given: "Agent Bell has completed language training."
Question: Is Bell a field agent?

Answer: Cannot determine. ✗

Why: "Most field agents are trained" does NOT mean "most trained people are field agents." Bell might be a field agent, or might not. The rule talks about the group, not individuals going the other direction.

Quick mental check: if you see "most" or "some" or "a few" in a premise and then a question about a SPECIFIC named person, the answer is almost always "cannot determine."

The 4-Step LBR Algorithm (use this on every question)

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STEP 1. Read the rule. Turn it into P → Q on scratch paper.
STEP 2. Write the CONTRAPOSITIVE: NOT Q → NOT P.
STEP 3. Look at what the question GIVES you as fact.
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STEP 4. Match against your P or NOT Q. Apply modus ponens
        or modus tollens. Nothing else is valid.
```

PART 2 — Detail Observation

They show you a picture for ~90 seconds. Then questions with the picture hidden.

The 90-Second Scan Pattern

Seconds	What to scan
0–15	Count PEOPLE (include uniformed staff and kids). Count vehicles. Note setting (lobby/street/etc).
15–45	Read EVERY number and word. Signs, plates, clocks, addresses. **Read plates TWICE.**
45–70	Per person: clothing COLOR, distinctive ITEM, direction of MOVEMENT, position.
70–90	Odd objects, unattended bags, red/bright items, cameras, exits.

Trap Examples

✗ TRAP **Trap 1 — Letter transposition on plates.**
Scene shows plate GTV-419.
Answer choices: (A) GVT-419 (B) GTV-419 (C) GTV-491
Options A and C each swap one detail. Option A flips the middle letters (T&V). Option C flips the last digits (91 vs 19).
Fix: Read plate TWICE during scan, say it out loud in your head.

✗ TRAP **Trap 2 — Object-color swap.**
Scene shows a woman in a GREEN dress carrying a RED purse.
Question asks: "What color was the woman's dress?"
Wrong answer: "Red" (you remember red but forget which object). Correct: "Green."
Fix: Always tag COLOR + OBJECT together, not color alone.

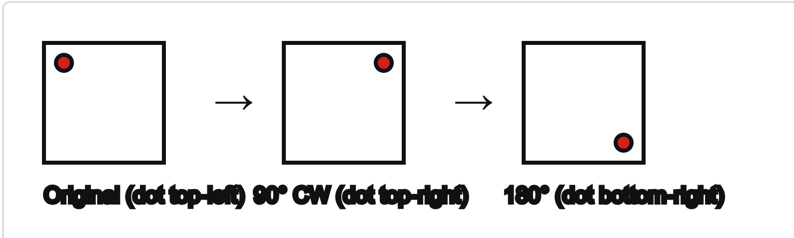
✗ TRAP **Trap 3 — "No X was visible" option.**
If you clearly remember X was there, never pick "X was not visible." The option exists to exploit doubt.

Two Scenes Back-to-Back (Interference)

> **When you study scene 2, your brain starts overwriting scene 1.** Counter-move: between scenes, spend 10 seconds mentally reciting scene 1's key facts (plate, count, colors) before starting scene 2.

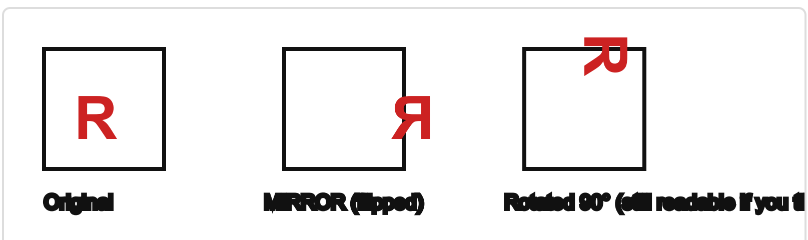
PART 3 — Figural Reasoning (Shape Puzzles)

Rotation: The Direction Map



90° clockwise: top-left → top-right → bottom-right → bottom-left → back to top-left.
180°: flips BOTH axes. Upside-down AND backward. The letter F becomes backwards and upside-down at the same time.
270° CW is the same as **90° counter-clockwise**.

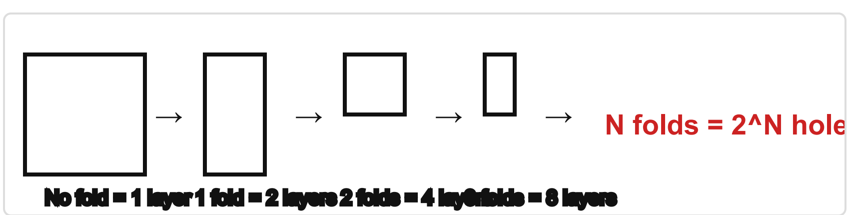
Rotation vs Mirror (the big one)



Rotation keeps the shape facing the same way — it just tilts it. A right-facing R stays right-facing (you just tilt your head to read it).
Mirror flips it. A right-facing R becomes backward.

Quick test: Could you get from the original to the answer by turning the paper? If yes = rotation. If you'd need to flip the paper over = mirror.

Paper Folding Math

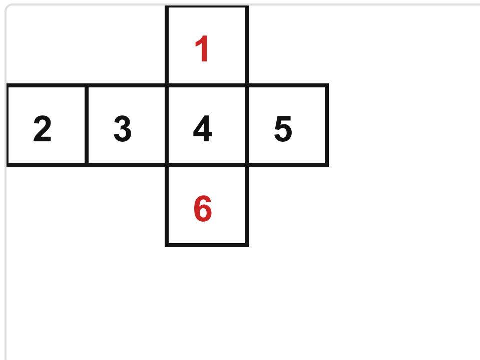


Simple formula: each fold doubles the layers. One hole punched through all layers makes one hole per layer when unfolded.

1 fold = 2 holes · 2 folds = 4 holes · 3 folds = 8 holes · 4 folds = 16 holes

Where are the holes? Holes are MIRROR-SYMMETRIC across every fold line. One hole near a fold gives two holes side-by-side when unfolded.

Cube Nets (flat shapes folded into cubes)



Two rules for opposites on a cross-shaped net:
· The top flap (1) is opposite the bottom flap (6).
· The two OUTER side flaps (2 and 5) are opposite each other.
· The CENTER face (3 or 4) is between them.

Shortcut: faces that touch along an edge on the flat net will NOT be opposite on the cube. Faces separated by exactly ONE face in a straight line ARE opposite.

Matrix Patterns (3×3 grid with a missing cell)

Always look for TWO rules:

- A ROW rule: what changes from left to right across each row? (shapes get more lines, dots increase, etc.)
- A COLUMN rule: what changes top to bottom in each column? (shape stays the same, rotation adds, etc.)

The missing cell combines BOTH rules. If you find only one rule, you're missing half the pattern.

Counting Shapes (like "how many triangles?")

Systematic count — not eyeball:

1. Count the smallest units first.
2. Count medium combinations (2 or 4 small units combined).
3. Count the biggest outer shape.
4. Add them up.

Example: A 3×3 grid has 14 squares total — 9 small (1×1) + 4 medium (2×2 overlapping) + 1 large (3×3).

PART 4 — Exam-Day Playbook

Night Before (Monday)

· Review this guide — 20 minutes max, NO new material
· Re-read the LBR translation table
· Lay out clothes, ID, pencils, water
· Bed EARLY. Sleep is the single biggest score multiplier

Tuesday Morning

· Breakfast: protein + complex carbs (eggs + oatmeal is ideal)
· Arrive 45 minutes early
· 5-minute walk before sitting down — oxygenates the brain
· Water, but not too much right before the exam starts

First 30 Seconds At Your Station

· Three slow breaths (4 in, 6 out)
· Read instructions FULLY — note timing per section
· If scratch paper is allowed, write the LBR translation table down IMMEDIATELY

Pacing Rules (stick to them)

LBR: 90 seconds per question MAXIMUM. Stuck? Eliminate obvious wrongs, guess, flag, move on.
Observation: 15 seconds per question. You either remember or you don't. Staring does not help.
Figural: 45 seconds per question (60 for counting questions).
NEVER leave a blank answer unless the exam confirms there's a guessing penalty (most civil-service exams don't have one).

If You Panic

· Skip brutal early questions — come back after easier wins
· Between sections: close eyes for 10 seconds, breathe, reset
· Feeling panic rise: press palms together HARD for 3 seconds (calms the vagus nerve)
· The clever-looking answer is usually the trap. The boring-looking answer is usually right.

PART 5 — Stare-At-This Cheat Card

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LBR:
  Translate rule to P → Q. Write the contrapositive NOT Q → NOT P.
  Only legal moves: modus ponens (P → Q) and modus tollens (NOT Q → NOT P).
  "Most / some / a few" = NO conclusion about individuals.
  "Only" flips the arrow. "Only X can Y" means Y → X.

OBSERVATION:
  Count people. Read plates TWICE. Tag color + object (not just color).
  Between scenes: rehearse scene 1 for 10 seconds before studying scene 2.

FIGURAL:
  Rotation keeps handedness. Mirror flips it.
  N folds = 2^N holes.
  Cube net: shared-edge faces are NEVER opposite.
  Matrix: find TWO rules (row + column).

META:
  Premises = absolute truth. Use only given info.
  Pick the WEAKEST true answer, not the strongest plausible.
  Contradictory premises = the answer IS "premises are inconsistent."
```

Top 10 Mistakes to Avoid

1. Affirming the consequent (P→Q, Q given, "so P" — wrong)
2. Denying the antecedent (P→Q, NOT P given, "so NOT Q" — wrong)
3. Using real-world logic instead of stated premises
4. Picking "strongest plausible" instead of "weakest true"
5. Treating "most" like "all"
6. Confusing rotation with mirror
7. Misreading a plate by one letter (GTV vs GVT)
8. Undercounting people (forgetting uniformed staff)
9. Burning too much time on one hard question
10. Trying to resolve a contradictory premise instead of naming it

You already nailed 90 practice questions. Tuesday is execution, not learning. Run the algorithms. Trust the process. You've got this.